End Semester Examination

Statistical Foundations for Machine Learning

December 2025

Information

- **Total Marks:** 50
- **Duration:** 3 hours
- There are two sections in the exam:
  - **Section A:** Short Answer Questions (5 questions × 2 marks = 10 marks)
  - **Section B:** Numerical Problems (8 questions × 5 marks = 40 marks)
- Write all answers in the answer sheet provided.
- Statistical tables (z-table, t-table, χ^2-table) are provided at the end.

Additional Instructions

- Answer all questions.
- Show all important steps and formulae used in your calculations.
- State assumptions clearly where necessary.
- Use appropriate notation throughout.

Section A: Short Answer Questions ($5 \times 2 = 10$ marks)

Q1. A sample of 100 observations has mean $\bar{x} = 45$ and standard deviation $s = 8$.

(a) What is the coefficient of variation for this sample?

(b) If each observation is transformed by $y_i = 2x_i + 5$, find the mean and standard deviation of the transformed data.

Solution to Q1.

(a) The coefficient of variation (CV) is

$$\text{CV} = \frac{s}{\bar{x}} \times 100\%.$$

$s = 8$ and $\bar{x} = 45$:

$$\text{CV} = \frac{8}{45} \times 100\% \approx 17.78\%.$$

(b) Each observation is transformed using

$$y_i = 2x_i + 5.$$

Mean of transformed data:

$$\bar{y} = 2\bar{x} + 5 = 2(45) + 5 = 95.$$

This file is meant for personal use by sjd1.contact@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Standard deviation of transformed data:

$$s_y = |2|s = 2(8) = 16.$$

Q2. Two events A and B satisfy: $P(A) = 0.6$, $P(B) = 0.4$, and $P(A \cap B) = 0.24$.

(a) Are A and B independent? Justify your answer.

(b) Find $P(A \cup B)$.

Solution to Q2.

(a) Events A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

$$P(A)P(B) = (0.6)(0.4) = 0.24.$$

Since

$$P(A \cap B) = 0.24 = P(A)P(B),$$

the events A and B are independent.

(b)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

$$P(A \cup B) = 0.6 + 0.4 - 0.24 = 0.76.$$

Q3. A continuous random variable X has probability density function

$$f(x) = \begin{cases} kx^2 & 0 \le x \le 2 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the value of constant k.

(b) Find $P(X > 1)$.

Solution to Q3.

(a) Since $f(x)$ is a pdf,

$$\int_0^2 kx^2 \, dx = 1.$$

$$k \int_0^2 x^2 \, dx = k \left[\frac{x^3}{3} \right]_0^2 = k \left(\frac{8}{3} \right) = 1.$$

$$k = \frac{3}{8}.$$

(b)

$$P(X > 1) = \int_1^2 kx^2 \, dx = \int_1^2 \frac{3}{8} x^2 \, dx.$$

$$P(X > 1) = \frac{3}{8} \left[\frac{x^3}{3} \right]_1^2 = \frac{3}{8} \left(\frac{8}{3} - \frac{1}{3} \right) = \frac{3}{8} \cdot \frac{7}{3} = \frac{7}{8}.$$

$$\boxed{P(X > 1) = \frac{7}{8}}$$

Q4. In a hypothesis test, the p-value is found to be 0.032.

This file is meant for personal use by sjd1.contact@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

(a) What decision would you make at significance level $\alpha = 0.05$?

(b) What decision would you make at significance level $\alpha = 0.01$?

Solution to Q4.
(a) Since the p-value is

$$0.032 < 0.05,$$

we reject the null hypothesis at the 5% significance level.

(b) Since

$$0.032 > 0.01,$$

we fail to reject the null hypothesis at the 1% significance level.

Q5. For a simple linear regression model $Y = \beta_0 + \beta_1 X + \epsilon$:

(a) If the coefficient of determination $R^2 = 0.85$, interpret this value.

(b) What does it mean if $\beta_1 = 0$?

Solution to Q5.
(a)

$$R^2 = 0.85$$

means that 85% of the variability in the response variable Y is explained by the linear relationship with the predictor variable X. The remaining 15% is due to other factors or random variation.

(b) If

$$\beta_1 = 0,$$

then the slope of the regression line is zero, meaning there is no linear relationship between X and Y. Changes in X do not predict changes in Y.

This file is meant for personal use by sjd1.contact@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Section B: Numerical Problems ($8 \times 5 = 40$ marks)

Q6. The following data represents the daily sales (in thousands of rupees) at a retail store over 12 days:

$$15,\ 18,\ 22,\ 25,\ 28,\ 30,\ 32,\ 35,\ 38,\ 42,\ 45,\ 50$$

(a) Calculate the mean and median of the sales data. [1 mark]

(b) Find the first quartile (Q_1) and third quartile (Q_3). [2 mark]

(c) Determine if there are any outliers using the IQR method. [2 mark].

Solution to Q6. Solution:
The given sales data (in thousands of rupees), arranged in ascending order, is:

$$15,\ 18,\ 22,\ 25,\ 28,\ 30,\ 32,\ 35,\ 38,\ 42,\ 45,\ 50$$

The number of observations is $n = 12$.

(a) **Mean and Median**

Mean:

$$\bar{x} = \frac{\sum x}{n}$$

$$\sum x = 15 + 18 + 22 + 25 + 28 + 30 + 32 + 35 + 38 + 42 + 45 + 50 = 380$$

$$\bar{x} = \frac{380}{12} = 31.67$$

Median:

Since $n = 12$ is even, the median is the average of the 6th and 7th observations.

$$\text{Median} = \frac{30 + 32}{2} = 31$$

$$\boxed{\text{Mean} = 31.67, \quad \text{Median} = 31}$$

—

(b) **First Quartile Q_1 and Third Quartile Q_3**

Lower half of the data:

$$15,\ 18,\ 22,\ 25,\ 28,\ 30$$

Upper half of the data:

$$32,\ 35,\ 38,\ 42,\ 45,\ 50$$

First Quartile Q_1:

$$Q_1 = \frac{22 + 25}{2} = 23.5$$

Third Quartile Q_3:

$$Q_3 = \frac{38 + 42}{2} = 40$$

$$\boxed{Q_1 = 23.5, \quad Q_3 = 40}$$

—

This file is meant for personal use by sjd1.contact@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

(c) **Outliers Using the IQR Method**

Interquartile Range:

$$\text{IQR} = Q_3 - Q_1 = 40 - 23.5 = 16.5$$

Lower and Upper Fences:

$$\text{Lower Fence} = Q_1 - 1.5(\text{IQR}) = 23.5 - 24.75 = -1.25$$

$$\text{Upper Fence} = Q_3 + 1.5(\text{IQR}) = 40 + 24.75 = 64.75$$

Since all data values lie between -1.25 and 64.75, there are no outliers.

No outliers are present in the dataset.

Q7. [Cricket Application] In a T20 cricket match, a batsman's probability of hitting a six on any given ball is 0.15, independent of other balls. In a particular over (6 balls), let X represent the number of sixes hit by the batsman.

(a) What is the probability distribution of X? State the parameters. [2 marks]

(b) Find the probability that the batsman hits exactly 2 sixes in the over. [1 mark]

(c) What is the expected number of sixes in the over? [1 mark]

(d) Find the variance of X. [1 mark]

Solution to Q7. Given: In a T20 cricket match, the probability of hitting a six on any given ball is

$$p = 0.15,$$

and the number of balls in one over is

$$n = 6.$$

Let X denote the number of sixes hit in one over.

—

(a) Probability Distribution of X:
Since each ball is an independent Bernoulli trial,

$$X \sim \text{Binomial}(n = 6,\, p = 0.15).$$

The probability mass function is:

$$P(X = k) = \binom{6}{k}(0.15)^k(0.85)^{6-k}, \quad k = 0, 1, 2, \ldots, 6.$$

—

(b) Probability of Exactly 2 Sixes:

$$P(X = 2) = \binom{6}{2}(0.15)^2(0.85)^4$$

$$= 15 \times 0.0225 \times 0.52200625$$

$$\approx 0.176.$$

—

This file is meant for personal use by sjd1.contact@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

(c) Expected Number of Sixes:

$$E(X) = np = 6 \times 0.15 = 0.9.$$

—

(d) Variance of X:

$$\text{Var}(X) = np(1-p)$$

$$= 6 \times 0.15 \times 0.85 = 0.765.$$

Q8. [**Financial Markets**] Daily returns (in percentage) of a stock over the past 5 trading days are:

$$2.1,\ -1.5,\ 3.2,\ -0.8,\ 1.4$$

Assume the population of daily returns is normally distributed with unknown variance.

(a) Calculate the sample mean return $\bar{x}$ and sample standard deviation s. [2 marks]

(b) Construct a 95% confidence interval for the true mean daily return μ. [2 marks]
(Use $t_{0.025,4} = 2.776$)

(c) Interpret your confidence interval in the context of stock returns. [1 mark].

Solution to Q8.

(a) **Sample mean return $\bar{x}$ and sample standard deviation s**

$$\bar{x} = \frac{\Sigma x}{5} = \frac{4.4}{5} = 0.88$$

and

$$(5-1)s^2 = \Sigma x^2 - 5\bar{x}^2 = 19.5 - 3.872 = 15.628$$

$$\implies s = \sqrt{\frac{15.628}{4}} = 1.9766$$

(b) **95% confidence interval for the true mean daily return μ**

Given:

$$n = 5,\ \bar{x} = 0.88,\ s = 1.9766,\ df = n - 1 = 4$$

.

$$Standard\ error : SE(\bar{X}) = \frac{s}{\sqrt{n}} = \frac{1.9766}{\sqrt{5}} = 0.88396$$

$$t_{0.025,4} = 2.776$$

95% CI for μ when variance unknown is

$$\bar{x} \pm t_{0.025,4} \times SE(\bar{X}) = 0.88 \pm 2.776 \times 0.88396 = 0.88 \pm 2.45387$$

95% CI for μ is

$$(-1.57387, 3.33387)$$

(c) **Interpretation of confidence interval in the context of stock returns**

We are 95% confident that the stock's true average daily return lies somewhere between -1.57% and $+3.33\%$. We observe higher inclination towards positive returns on average but still there can be negative returns on average.

This file is meant for personal use by sjd1.contact@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Q9. A quality control inspector wants to test whether the mean weight of cereal boxes is 500 grams. A random sample of 25 boxes yields $\bar{x} = 497$ grams with sample standard deviation $s = 12$ grams. Assume the population is normally distributed.

(a) State the null and alternative hypotheses for a two-sided test. [1 mark].

(b) Calculate the test statistic. [2 marks]

(c) At significance level $\alpha = 0.05$, make a decision regarding H_0. [1 mark]
(Use $t_{0.025,24} = 2.064$)

(d) What is your conclusion in the context of the problem? [1 mark]

Solution to Q9.
We are testing a claim about the mean weight μ of cereal boxes, based on a sample of size $n = 25$ with:

$$\bar{x} = 497, \quad s = 12, \quad n = 25.$$

(a) **Hypotheses for this two-sided test.**

$$H_0 : \mu = 500 \quad \text{(the mean weight } \mu \text{ of cereal boxes is 500 grams)}$$
$$H_1 : \mu \neq 500 \quad \text{(the mean weight } \mu \text{ of cereal boxes is not 500 grams)}$$

This is a **two-tailed** t-test.

(b) **The test statistic.**

Since the population standard deviation is unknown and the population is assumed normal, we use the t-statistic:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}},$$

where $\mu_0 = 500$ is the hypothesized mean.

Substitute the values:

$$t = \frac{497 - 500}{12/\sqrt{25}} = \frac{-3}{12/5} = \frac{-5}{4} = -1.25$$

The test statistic is:

$$t_{\text{obs}} = -1.25.$$

(c) **Decision regarding H_0.**

The degrees of freedom are:

$$\text{df} = n - 1 = 25 - 1 = 24.$$

$$\alpha = 0.05 \implies \alpha/2 = 0.025.$$

$$t_{0.025,24} = 2.064$$

Hence, our critical/rejection region is:

$$(-\infty, -2.064) \cup (2.064, \infty)$$

And we observe $t_{\text{obs}} = -1.25$, which do not belong to the critical region.
Hence, we **fail to reject** H_0 at the 5% level.

This file is meant for personal use by sjd1.contact@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

(d) **Conclusion.**
There is insufficient statistical evidence at the 5% significance level to reject the inspector's claim that the true mean weight of cereal boxes is 500 grams. In other words, the sample data are consistent with the machine having an average box weight of 500 grams.

Q10. Let X and Y be jointly distributed discrete random variables with the following joint probability mass function:

$X \backslash Y$	0	1	2	Total
1	0.10	0.15	0.05	0.30
2	0.20	0.10	0.15	0.45
3	0.05	0.10	0.10	0.25
Total	0.35	0.35	0.30	1.00

(a) Find the marginal distributions of X and Y. [2 marks]

(b) Calculate $E[X]$ and $E[Y]$. [2 marks]

(c) Are X and Y independent? Justify your answer with appropriate calculations. [1 mark]

Solution to Q10.

(a) **Marginal Distributions:**

$$P(X = 1) = 0.30, \quad P(X = 2) = 0.45, \quad P(X = 3) = 0.25.$$

$$P(Y = 0) = 0.35, \quad P(Y = 1) = 0.35, \quad P(Y = 2) = 0.30.$$

(b) **Expected Values:**

$$E[X] = 1(0.30) + 2(0.45) + 3(0.25) = 0.30 + 0.90 + 0.75 = 1.95.$$

$$E[Y] = 0(0.35) + 1(0.35) + 2(0.30) = 0 + 0.35 + 0.60 = 0.95.$$

(c) **Independence Check:**

For $(X, Y) = (1, 0)$:

$$P(X = 1, Y = 0) = 0.10,$$
$$P(X = 1)P(Y = 0) = (0.30)(0.35) = 0.105.$$

Since

$$P(X = 1, Y = 0) \neq P(X = 1)P(Y = 0),$$

we conclude that X **and** Y **are not independent.**

Q11. A study was conducted to examine the relationship between hours of study per week (X) and exam scores (Y) for 6 students. The following data was collected:

X (hours)	5	10	15	20	25	30
Y (scores)	45	55	65	75	80	90

Given: $\sum x_i = 105$, $\sum y_i = 410$, $\sum x_i^2 = 2275$, $\sum y_i^2 = 29350$, $\sum x_i y_i = 7775$

(a) Calculate the sample correlation coefficient r. [1 mark]

This file is meant for personal use by sjd1.contact@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

(b) Find the least squares regression line $\hat{y} = b_0 + b_1 x$. [2 marks]

(c) Interpret the slope b_1 in the context of the problem. [1 mark]

(d) Predict the exam score for a student who studies 18 hours per week. [1 mark].

Solution to Q11.

Given:

$$n = 6, \quad \sum x_i = 105, \quad \sum y_i = 410, \quad \sum x_i^2 = 2275, \quad \sum y_i^2 = 29350, \quad \sum x_i y_i = 7775$$

Mean values:

$$\bar{x} = \frac{105}{6} = 17.5, \qquad \bar{y} = \frac{410}{6} = 68.33$$

$$S_{xx} = \sum x_i^2 - n\bar{x}^2 = 2275 - 6(17.5)^2 = 437.5$$

$$S_{yy} = \sum y_i^2 - n\bar{y}^2 = 29350 - 6(68.33)^2 = 1333.33$$

$$S_{xy} = \sum x_i y_i - n\bar{x}\bar{y} = 7775 - 6(17.5)(68.33) = 600$$

(a) Sample correlation coefficient:

$$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}} = \frac{600}{\sqrt{(437.5)(1333.33)}} \approx 0.786$$

(b) Least squares regression line:

$$b_1 = \frac{S_{xy}}{S_{xx}} = \frac{600}{437.5} = 1.371$$

$$b_0 = \bar{y} - b_1 \bar{x} = 68.33 - (1.371)(17.5) = 44.33$$

$$\boxed{\hat{y} = 44.33 + 1.371x}$$

(c) Interpretation of slope:

For every additional hour of study per week, the exam score increases on average by approximately **1.37 marks**.

(d) Prediction for $x = 18$:

$$\hat{y} = 44.33 + 1.371(18) = 69.02$$

$$\boxed{\hat{y}(18) \approx 69.02}$$

Alternate

This file is meant for personal use by sjd1.contact@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Solution using the supplied sums

Given:

$$n = 6, \qquad \sum x = 105, \qquad \sum y = 410, \qquad \sum x^2 = 2275, \qquad \sum y^2 = 29400, \qquad \sum xy = 7950.$$

(a) **Sample correlation coefficient r.**

Using

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{\left[n\sum x^2 - (\sum x)^2\right]\left[n\sum y^2 - (\sum y)^2\right]}}.$$

Compute:

$$\text{Numerator} = 6(7950) - 105 \cdot 410 = 47700 - 43050 = 4650,$$

$$n\sum x^2 - (\sum x)^2 = 6(2275) - 105^2 = 13650 - 11025 = 2625,$$

$$n\sum y^2 - (\sum y)^2 = 6(29400) - 410^2 = 176400 - 168100 = 8300.$$

Hence

$$r = \frac{4650}{\sqrt{2625 \cdot 8300}} = \frac{4650}{4668.886\ldots} \approx 0.996206.$$

$$\boxed{r \approx 0.996206}$$

(b) **Least squares regression line $\hat{y} = b_0 + b_1 x$.**

Use summation formulas:

$$b_1 = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}, \qquad b_0 = \bar{y} - b_1\bar{x},$$

with

$$\bar{x} = \frac{105}{6} = 17.5, \qquad \bar{y} = \frac{410}{6} = 68.333\ldots$$

Compute slope and intercept:

$$b_1 = \frac{4650}{2625} = 1.7714285714\ldots, \qquad b_0 = 68.333\ldots - 1.7714285714(17.5) = 37.3333333\ldots$$

Thus

$$\boxed{\hat{y} = 37.3333 + 1.77143\,x.}$$

(c) **Interpretation of b_1.**

$b_1 \approx 1.7714$ means: according to this fitted model (using the supplied sums), each additional hour of study per week is associated with an average increase of approximately **1.77** exam-score points.

(d) **Prediction for $x = 18$.**

$$\hat{y}(18) = 37.3333333 + 1.7714285714 \times 18 \approx 69.21905.$$

$$\boxed{\hat{y}(18) \approx 69.219}$$

This file is meant for personal use by sjd1.contact@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Q12. A pharmaceutical company claims that a new drug reduces recovery time with mean recovery time of 10 days. A random sample of 16 patients shows a mean recovery time of 11.5 days with standard deviation of 3 days. A researcher believes the actual mean recovery time is higher than claimed. Assume normal population.

(a) State appropriate hypotheses for this one-sided test. [1 mark]

(b) Calculate the test statistic. [2 marks]

(c) Find the p-value for your test statistic. (Use $t_{0.05,15} = 1.753$, $t_{0.025,15} = 2.131$) [1 mark]

(d) At $\alpha = 0.05$, what is your conclusion? Justify. [1 mark].

Solution to Q12.

We are testing a claim about the population mean recovery time μ of the drug, based on a sample of size $n = 16$ with:

$$\bar{x} = 11.5, \quad s = 3, \quad n = 16.$$

(a) **State appropriate hypotheses for this one-sided test.**

The company claims the mean recovery time is 10 days. The researcher believes the *actual* mean is *higher* than claimed.

Therefore:

$$H_0 : \mu = 10 \quad \text{(the mean recovery time is 10 days)}$$

$$H_1 : \mu > 10 \quad \text{(the mean recovery time is greater than 10 days)}$$

This is a **right-tailed** (upper-tailed) one-sample t-test.

(b) **Calculate the test statistic.**

Since the population standard deviation is unknown and the population is assumed normal, we use the one-sample t-statistic:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}},$$

where $\mu_0 = 10$ is the hypothesized mean.

Substitute the values:

$$t = \frac{11.5 - 10}{3/\sqrt{16}} = \frac{1.5}{3/4} = \frac{1.5}{0.75} = 2.0.$$

The test statistic is:

$$t_{\text{obs}} = 2.0.$$

The degrees of freedom are:

$$\text{df} = n - 1 = 16 - 1 = 15.$$

(c) **Find the p-value for your test statistic.**

We need the p-value for a right-tailed t-test:

$$\text{p-value} = P(T \geq t_{\text{obs}}) = P(T \geq 2.0),$$

where T follows a t-distribution with $\text{df} = 15$.

We are given:

$$t_{0.05,15} = 1.753, \quad t_{0.025,15} = 2.131,$$

This file is meant for personal use by sjd1.contact@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

where $t_{\alpha,15}$ denotes the critical value such that

$$P(T \geq t_{\alpha,15}) = \alpha \quad \text{for } T \sim t_{15}.$$

We compare our observed t:

$$1.753 < 2.0 < 2.131.$$

This implies:

$$P(T \geq 1.753) = 0.05, \quad P(T \geq 2.131) = 0.025,$$

and since 2.0 lies between these two critical values,

$$0.025 < P(T \geq 2.0) < 0.05.$$

Therefore, the p-value satisfies:

$$0.025 < \text{p-value} < 0.05.$$

(If an approximate numerical value is desired, one could state that the p-value is around 0.03–0.04, but the interval above is sufficient using the given table values.)

(d) **At $\alpha = 0.05$, what is your conclusion? Justify.**

The decision rule using the p-value is:

- If p-value $\leq \alpha$, reject H_0.
- If p-value $> \alpha$, fail to reject H_0.

From part (c), we found:

$$0.025 < \text{p-value} < 0.05,$$

so in particular:

$$\text{p-value} < 0.05 = \alpha.$$

Hence, at significance level $\alpha = 0.05$, we **reject the null hypothesis** H_0.

Conclusion (in context): There is sufficient statistical evidence at the 5% significance level to support the researcher's claim that the true mean recovery time is *higher* than the company's claimed 10 days.

Q13. (Bayes' Theorem and Law of Total Probability)

A large company conducts an annual health check-up for its employees. Each employee is first classified as either *high-risk* or *low-risk* based on lifestyle and medical history.

- Let F be the event that a randomly chosen employee is *high-risk*. It is known that $P(F) = 0.20$.
- The remaining employees are classified as *low-risk*, so F^c is the event "employee is low-risk".
- Let E be the event that an employee *has a certain disease* detected by the screening.

Medical data from previous years suggest that

$$P(E \mid F) = 0.12 \quad \text{and} \quad P(E \mid F^c) = 0.03.$$

This file is meant for personal use by sjd1.contact@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

(a) Using the identity

$$P(E) = P(E \mid F)P(F) + P(E \mid F^c)\big[1 - P(F)\big],$$

write $P(E)$ explicitly in terms of the given numerical values. Do *not* simplify yet; just substitute. [1 mark]

(b) Compute the numerical value of $P(E)$, the overall probability that a randomly chosen employee has the disease. [1 mark]

(c) Using Bayes' Theorem, compute the probability that an employee is *high-risk* given that they have the disease, i.e., find $P(F \mid E)$. [1 mark]

(d) Interpret your result in part (c) in words. In particular, compare $P(F)$ (the prior probability of being high-risk) with $P(F \mid E)$ (the probability of being high-risk among those who actually have the disease). What does this tell you about the relationship between being high-risk and having the disease? [2 marks].

Solution.

(a) We are given:

$$P(F) = 0.20, \quad P(F^c) = 1 - P(F) = 0.80,$$
$$P(E \mid F) = 0.12, \quad P(E \mid F^c) = 0.03.$$

Using the given identity

$$P(E) = P(E \mid F)P(F) + P(E \mid F^c)\big[1 - P(F)\big],$$

we substitute the values:

$$P(E) = (0.12)(0.20) + (0.03)(0.80).$$

(b) Now we compute $P(E)$:

$$(0.12)(0.20) = 0.024, \qquad (0.03)(0.80) = 0.024.$$

Hence,

$$P(E) = 0.024 + 0.024 = 0.048.$$

So the overall probability that a randomly chosen employee has the disease is

$$\boxed{P(E) = 0.048}.$$

(c) By Bayes' Theorem, the probability that an employee is high-risk given that they have the disease is

$$P(F \mid E) = \frac{P(E \cap F)}{P(E)} = \frac{P(E \mid F)P(F)}{P(E)}.$$

Substituting the values:

$$P(F \mid E) = \frac{(0.12)(0.20)}{0.048} = \frac{0.024}{0.048} = 0.5.$$

Thus

$$\boxed{P(F \mid E) = 0.50}.$$

This file is meant for personal use by sjd1.contact@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

(d) Interpretation:

Originally, before knowing anything about the disease status of an employee, the probability that a randomly chosen employee is high-risk is

$$P(F) = 0.20 \quad (20\%).$$

However, among the employees who *do* have the disease, the probability of being high-risk is

$$P(F \mid E) = 0.50 \quad (50\%).$$

This means that, although only 20% of all employees are high-risk, they make up 50% of the employees who actually have the disease. In other words, knowing that an employee has the disease *raises* the likelihood that they belong to the high-risk group. This illustrates how Bayes' Theorem updates our belief about F (high-risk) once we condition on the evidence E (presence of disease).

— End of Question Paper —

This file is meant for personal use by sjd1.contact@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.